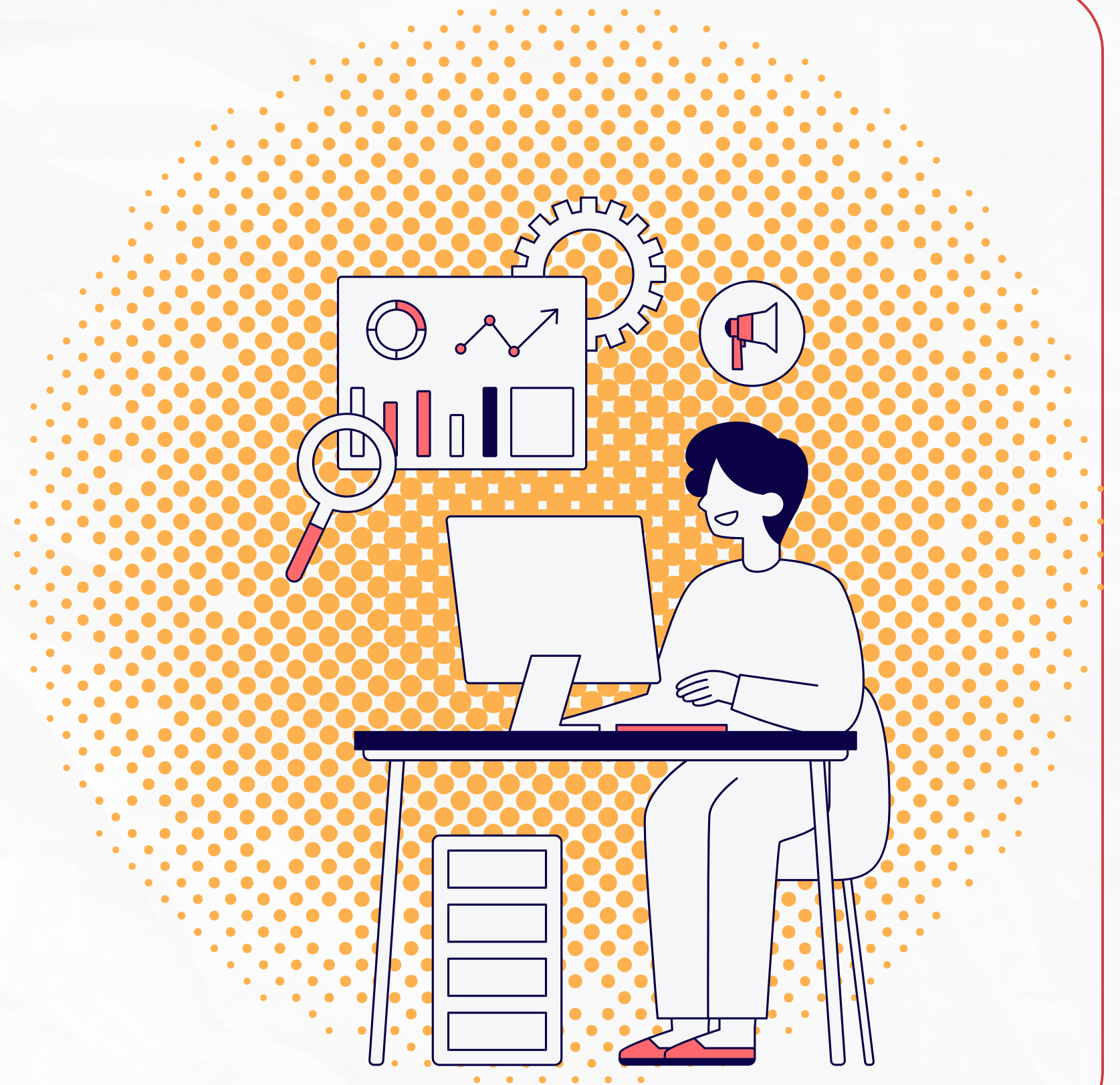


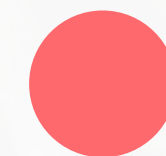
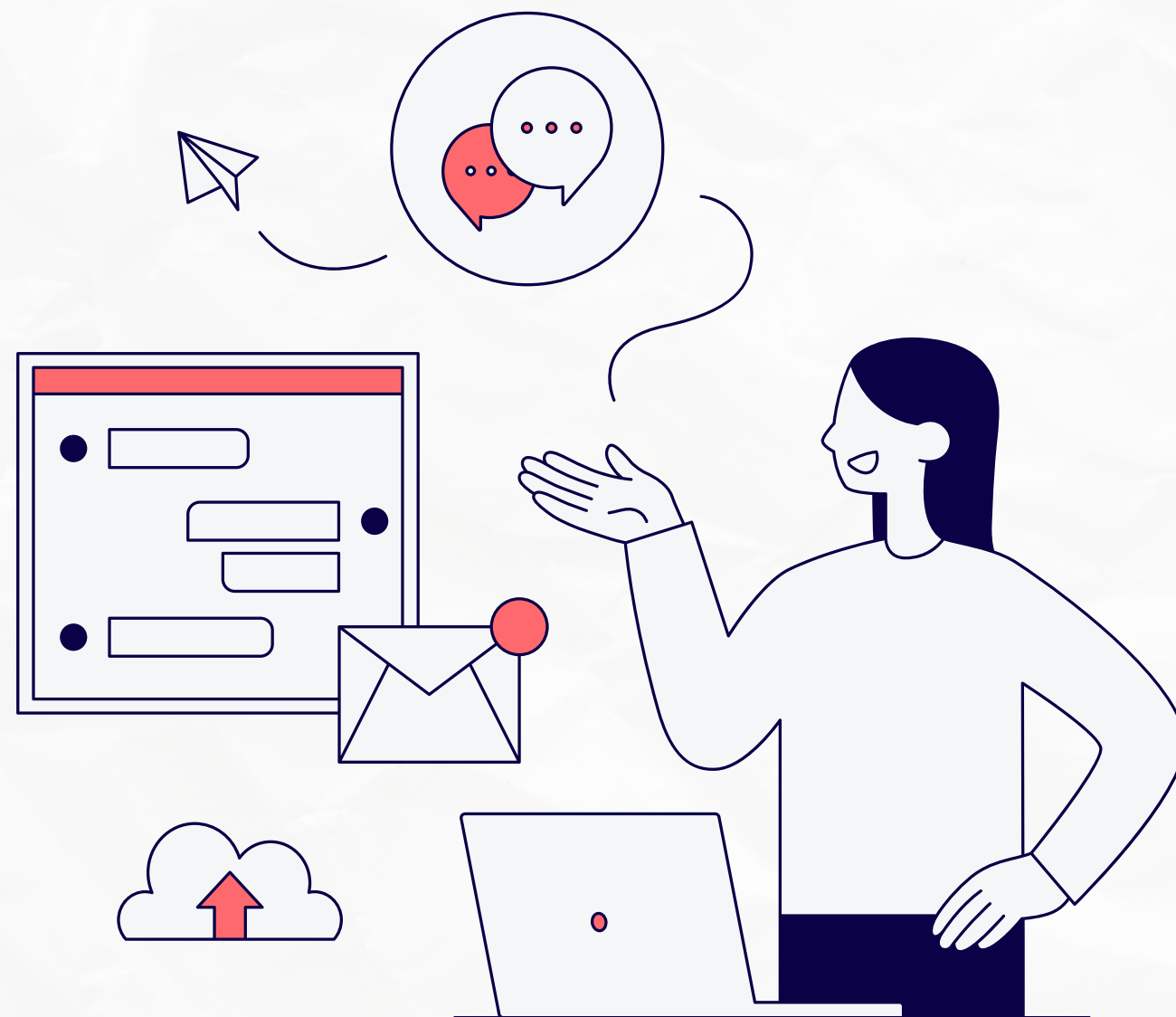
FIFA RANKINGS 1993-2018

DATA ANALYSIS

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About the Dataset



FIFA Soccer Rankings International Men's Ranking (August 1993 – June 2018).

Each year since 1992 FIFA has released rankings on all international teams thorough an elo system. This dataset contains the rankings of each international team from 1993–2018 for men's soccer. The information from the dataset was taken from FIFA's official website.

Which league is better?

The main two federations that have been the best in football for a long time have been UEFA and CONMEBOL corresponding to Europe and South America respectively. Currently all of the top 10 teams are from either of these two leagues. The question that we are asking is if there is a significant difference between the rankings for each of the leagues as well as if so which league is better.



Hypothesis Testing

1

Hypothesis: After controlling for the number of matches played, UEFA and CONMEBOL teams have significantly different average FIFA ranking points.

2

Null Hypothesis: After controlling for the number of matches played, there is no difference in average FIFA ranking points between UEFA teams and CONMEBOL teams.

Statistical Summary

1

***Data tested in Rstudio using column:
"previous_points"



2

- Mean for UEFA: 501.9545
- Mean for CONMEBOL: 636.436
- Difference in means: -134.4815
- Z-score: -16.65341
- P-value: 0

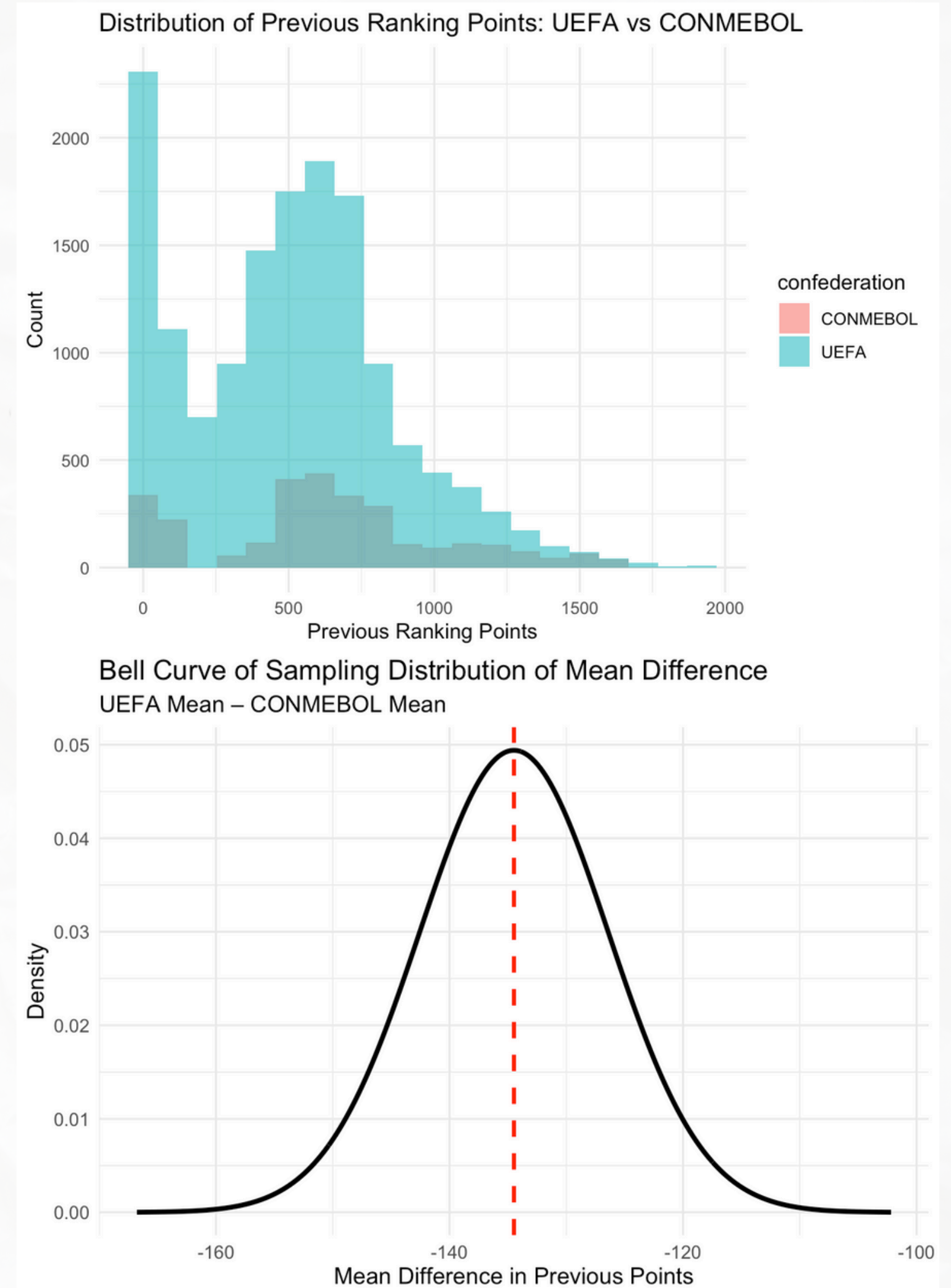
Analysis + Graph

1

Analysis of Statistical Data (previous slide)

The mean previous ranking points for UEFA teams (501.95) are substantially lower than those for CONMEBOL teams (636.44). The difference in means is -134.48 , indicating that CONMEBOL teams outperform UEFA teams by a wide margin on this metric.

The Z-score of -16.65 is extremely large in magnitude, and the corresponding p-value is effectively 0, well below any conventional significance threshold. This provides overwhelming evidence that the difference is not due to random sampling variation and the null is **rejected**.



Hypothesis Testing - Rscript

```
# --- 1. Subset Data (assumes data already imported as fifa_ranking) ---
df <- fifa_ranking

df_sub <- subset(df, confederation %in% c("UEFA", "CONMEBOL"))
df_sub$confederation <- factor(df_sub$confederation)

# --- 2. Convert previous_points to numeric ---
df_sub$previous_points <- as.numeric(gsub("[^0-9.]", "", df_sub$previous_points))

# --- 3. Compute Means ---
mean_uefa <- mean(df_sub$previous_points[df_sub$confederation == "UEFA"], na.rm = TRUE)
mean_conmebol <- mean(df_sub$previous_points[df_sub$confederation == "CONMEBOL"], na.rm = TRUE)

mean_uefa
mean_conmebol

# --- 4. Difference in Means ---
diff_means <- mean_uefa - mean_conmebol
diff_means

# --- 5. Compute Standard Error and Z-score ---
n_uefa <- sum(df_sub$confederation == "UEFA")
n_conmebol <- sum(df_sub$confederation == "CONMEBOL")

sd_uefa <- sd(df_sub$previous_points[df_sub$confederation == "UEFA"], na.rm = TRUE)
sd_conmebol <- sd(df_sub$previous_points[df_sub$confederation == "CONMEBOL"], na.rm = TRUE)

se_diff <- sqrt((sd_uefa^2 / n_uefa) + (sd_conmebol^2 / n_conmebol))

z_score <- diff_means / se_diff
z_score

# --- 6. Two-tailed P-value ---
p_value <- 2 * (1 - pnorm(abs(z_score)))
p_value

# --- 7. Graphs Supporting the Hypothesis Test ---
library(ggplot2)

## Histogram comparing distributions
ggplot(df_sub, aes(x = previous_points, fill = confederation)) +
  geom_histogram(alpha = 0.6, position = "identity", bins = 20) +
  labs(title = "Distribution of Previous Ranking Points: UEFA vs CONMEBOL",
       x = "Previous Ranking Points", y = "Count") +
  theme_minimal()
```

```
## Boxplot comparison
ggplot(df_sub, aes(x = confederation, y = previous_points, fill = confederation)) +
  geom_boxplot(alpha = 0.8) +
  labs(title = "Previous FIFA Ranking Points by Confederation",
       x = "", y = "Previous Ranking Points") +
  theme_minimal()

## Mean comparison plot
mean_df <- data.frame(
  Confederation = c("UEFA", "CONMEBOL"),
  MeanPoints = c(mean_uefa, mean_conmebol)
)

ggplot(mean_df, aes(x = Confederation, y = MeanPoints, fill = Confederation)) +
  geom_col(alpha = 0.85) +
  geom_text(aes(label = round(MeanPoints, 1)), vjust = -0.5, size = 5) +
  labs(title = "Mean Previous Ranking Points: UEFA vs CONMEBOL",
       x = "", y = "Mean Previous Ranking Points") +
  theme_minimal()

# --- 10. Bell Curve for Sampling Distribution of Mean Difference ---

# Difference in means (already computed)
diff_means

# Standard error of difference (already computed)
se_diff

# Create x-values centered around the observed difference
x_vals <- seq(diff_means - 4 * se_diff,
              diff_means + 4 * se_diff,
              length.out = 500)
```

```
library(ggplot2)

ggplot(bell_diff_df, aes(x = x, y = density)) +
  geom_line(size = 1.2) +
  geom_vline(xintercept = diff_means,
             color = "red", linetype = "dashed", linewidth = 1) +
  labs(
    title = "Bell Curve of Sampling Distribution of Mean Difference",
    subtitle = "UEFA Mean - CONMEBOL Mean",
    x = "Mean Difference in Previous Points",
    y = "Density"
  ) +
  theme_minimal() +
  theme(text = element_text(size = 12))
```




Hypothesis Conclusion



There is a statistically significant and meaningful difference in previous FIFA ranking points between UEFA and CONMEBOL, with CONMEBOL teams showing much higher average performance.

Data Exploration and Visualization Tools

1

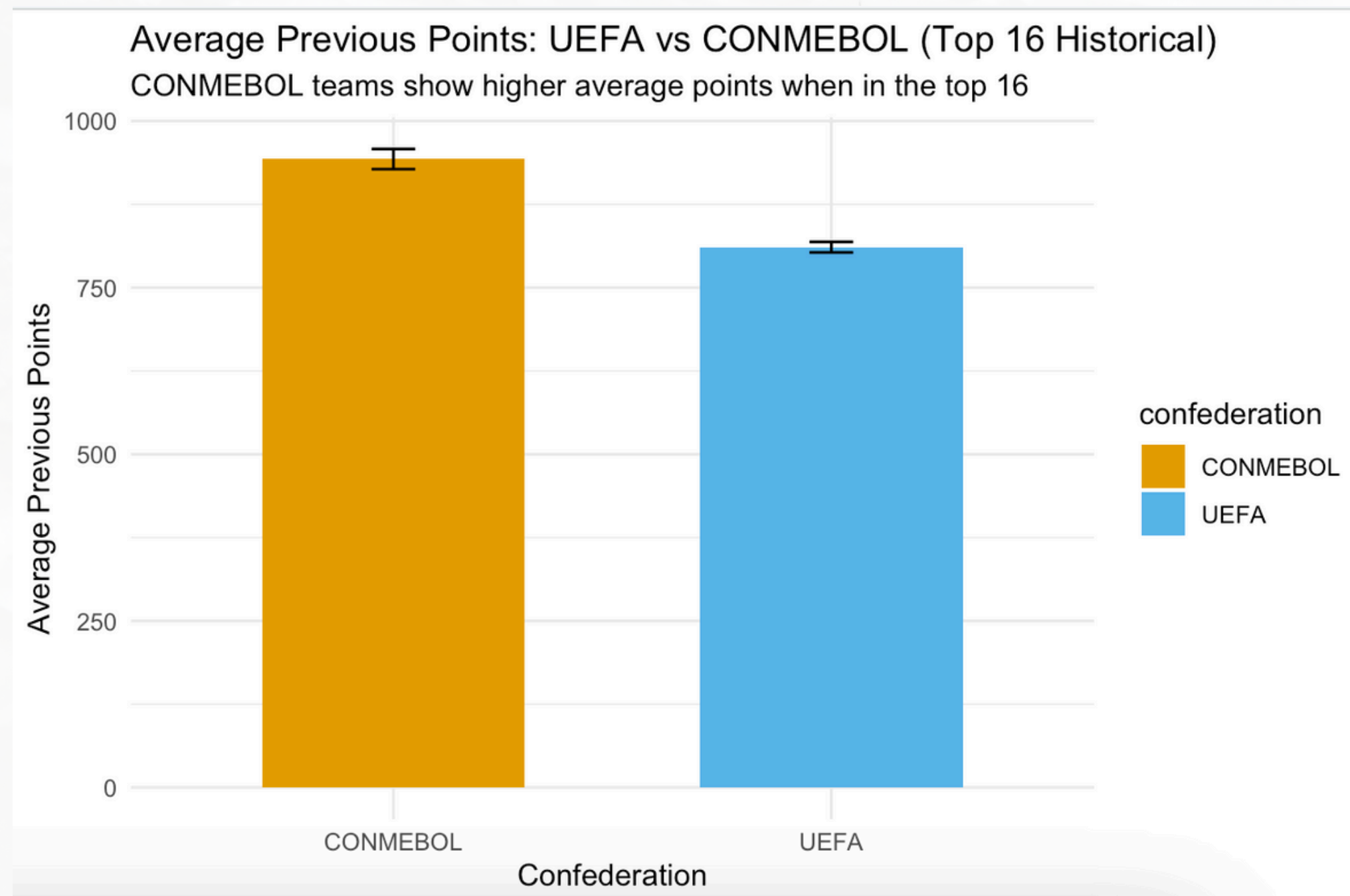
We will use a bar graph of the top 16 countries that shows the significantly higher average points for CONMEBOL across all historical top 16 countries.

2

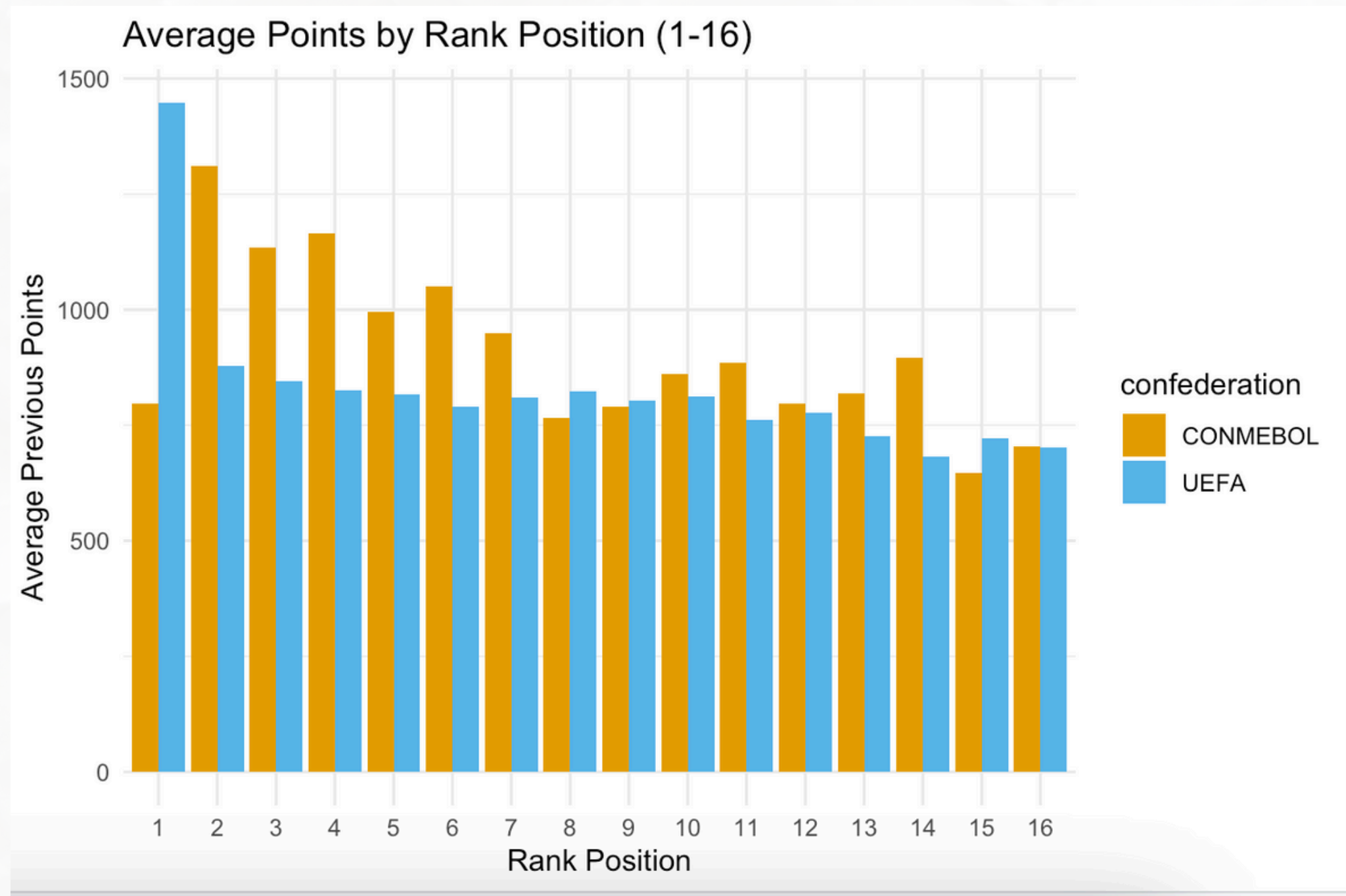
This demonstrates that CONMEBOL teams consistently hold higher point averages across nearly every specific rank position from 1 to 16.



Bar Graph for Average Previous Points



Bar Graph for Average Points By Rank




```

1 library(ggplot2)
2 library(dplyr)
3
4 fifa_data <- read.csv("fifa_ranking.csv")
5
6 top_16_data <- fifa_data %>%
7   filter(rank <= 16) %>%
8   filter(confederation %in% c("UEFA", "CONMEBOL"))
9
10 print(aggregate(previous_points ~ confederation, top_16_data, mean))
11
12 p1 <- ggplot(top_16_data, aes(x = confederation, y = previous_points, fill = confederation)) +
13   geom_bar(stat = "summary", fun = "mean", width = 0.6) +
14   stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
15   scale_fill_manual(values = c("CONMEBOL" = "#E69F00", "UEFA" = "#56B4E9")) +
16   labs(title = "Average Previous Points: UEFA vs CONMEBOL (Top 16 Historical)",
17        subtitle = "CONMEBOL teams show higher average points when in the top 16",
18        y = "Average Previous Points",
19        x = "Confederation") +
20   theme_minimal()
21
22 p2 <- ggplot(top_16_data, aes(x = factor(rank), y = previous_points, fill = confederation)) +
23   geom_bar(stat = "summary", fun = "mean", position = "dodge") +
24   scale_fill_manual(values = c("CONMEBOL" = "#E69F00", "UEFA" = "#56B4E9")) +
25   labs(title = "Average Points by Rank Position (1-16)",
26        y = "Average Previous Points",
27        x = "Rank Position") +
28   theme_minimal()
29 |
30 print(p1)
31 print(p2)

```

Conclusion from the Graph

Based on the analysis of the dataset, there is indeed a statistically significant difference when examining the historical performance of teams within the top 16 ranking positions.

Historically, CONMEBOL teams have maintained a higher average of previous ranking points (approx. 943) compared to UEFA teams (approx. 811) when occupying these top spots ($p < 0.001$).

While UEFA has a higher volume of teams, the density of quality at the top is higher for CONMEBOL. This means that is a South American team entering the top 16 is statistically likely to be a stronger team than a European team if they were at the same rank.